E-Commerce Database Management System

Task 2: Product and Category Management

1. Category and Product Table Design

Task 2 separates product categories from the product table. Instead of typing a category name against every product, each product stores a category_id that points to a row in a new category table. This avoids spelling mistakes and repeated text, and a category's description can be changed in one place.

Task 2 objectives

- Create the category and product tables with primary key, foreign key, NOT NULL, UNIQUE and CHECK constraints.
- Insert category records and at least 10 product records.
- Perform INSERT, SELECT, UPDATE and DELETE operations.
- Run category-wise queries: products by category, product count, highest price, categories with more than 5 products, and average price.

1.1 Category (table: category)

Attribute	Data type	Constraint	Description
category_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID of the category
category_name	VARCHAR(50)	NOT NULL, UNIQUE, CHECK (name is not blank)	Name of the category
description	VARCHAR(255)	Optional	Short description of the category

1.2 Product (table: product)

Attribute	Data type	Constraint	Description
product_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID of the product
product_name	VARCHAR(150)	NOT NULL	Name of the product
category_id	INT	NOT NULL, FOREIGN KEY references category(category_id)	Category the product belongs to

Attribute	Data type	Constraint	Description
price	DECIMAL(10,2)	NOT NULL, CHECK (price > 0)	Selling price
stock_quantity	INT	NOT NULL, DEFAULT 0, CHECK (stock_quantity >= 0)	Units available
seller_id	INT	FOREIGN KEY references seller(seller_id)	Seller who supplies the product (from Task 1, used again in Week 3)

1.3 Constraint Summary

Constraint	Purpose	Where it is used
PRIMARY KEY	Identifies each row uniquely	category.category_id, product.product_id
FOREIGN KEY	Links a row to a row in another table	product.category_id to category, product.seller_id to seller
NOT NULL	Makes a value compulsory	category_name, product_name, category_id, price, stock_quantity
UNIQUE	Prevents duplicate values	category.category_name
CHECK	Rejects invalid values	category_name is not blank, price > 0, stock_quantity >= 0
DEFAULT	Supplies a value when none is given	stock_quantity (0)

1.4 Relationship Between Category and Product

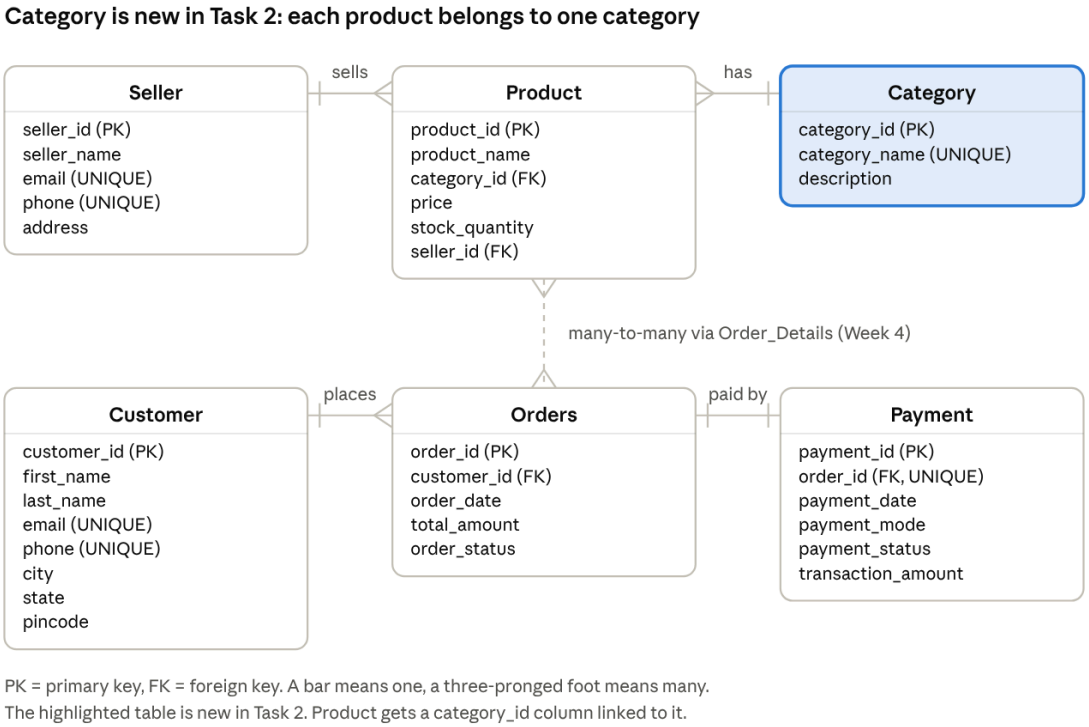
Relationship	Type	Linked by	Explanation
Category to Product	One-to-Many	product.category_id	One category contains many products. Each product belongs to exactly one category.

Referential integrity rules

- A product cannot be added with a category that does not exist.
- A category that still has products cannot be deleted (MySQL error 1451). Its products must be deleted or moved first.
- Category names cannot repeat, so the same category is never stored twice.

2. ER Diagram

The updated ER diagram adds the category table to the five entities from Task 1. The highlighted table is new in Task 2, and the crow's foot marks the many side of a relationship.



Updated ER diagram · 6 entities, 5 relationships

Each category contains many products, and each product belongs to exactly one category. The relationships from Task 1 are unchanged.

3. Sample Product Data

3.1 Category data

Eight categories are inserted. Their IDs are assigned automatically in this order.

Category_ID	Category_Name	Description
1	Electronics	Mobiles, audio devices and accessories
2	Fashion	Clothing and footwear for men and women
3	Home & Kitchen	Cookware, appliances and household items
4	Books	Fiction, non-fiction and self-help books

Category_ID	Category_Name	Description
5	Sports	Sports equipment and fitness gear
6	Beauty	Skin care and personal care products
7	Toys	Toys and games for children
8	Grocery	Packaged food and organic groceries

3.2 Product data

Products 1 to 10 come from Task 1 and are assigned categories with UPDATE statements. Products 11 to 20 are new. Electronics has 6 products so that the query for categories with more than 5 products returns a result.

Product_ID	Product_Name	Category_ID	Category	Price	Stock_Quantity
1	Redmi Note 13 5G	1	Electronics	16999.00	50
2	Men Cotton T-Shirt	2	Fashion	499.00	200
3	Non-stick Cookware Set	3	Home & Kitchen	1899.00	80
4	Wings of Fire (Book)	4	Books	250.00	150
5	Kashmir Willow Cricket Bat	5	Sports	1299.00	40
6	Wireless Headphones	1	Electronics	1499.00	120
7	Mixer Grinder 750W	3	Home & Kitchen	3299.00	60
8	Face Wash 100ml	6	Beauty	199.00	300
9	Building Blocks Set	7	Toys	799.00	90
10	Organic Honey 500g	8	Grocery	350.00	110
11	Bluetooth Speaker	1	Electronics	1999.00	70
12	Smart Watch	1	Electronics	2499.00	45
13	Power Bank 20000mAh	1	Electronics	1299.00	100

Product_ID	Product_Name	Category_ID	Category	Price	Stock_Quantity
14	USB-C Fast Charger 65W	1	Electronics	999.00	85
15	Women Kurti	2	Fashion	699.00	120
16	Running Shoes	2	Fashion	2299.00	60
17	Steel Water Bottle 1L	3	Home & Kitchen	399.00	150
18	Atomic Habits (Book)	4	Books	499.00	90
19	Yoga Mat	5	Sports	699.00	75
20	Moisturizer 200ml	6	Beauty	349.00	140

The price of product 1 is 16999.00 because it was reduced from 17999.00 by an UPDATE in Task 1. In the CRUD section the price of product 11 is changed to 1799.00 and the stock of product 12 is raised from 45 to 70. The analysis report below uses the data after these updates.

4. Category-wise Analysis Report

This report presents the results of the category-wise queries in the SQL script, followed by the main observations.

4.1 Products displayed category-wise

Category	Products
Beauty	Face Wash 100ml, Moisturizer 200ml
Books	Atomic Habits (Book), Wings of Fire (Book)
Electronics	Bluetooth Speaker, Power Bank 20000mAh, Redmi Note 13 5G, Smart Watch, USB-C Fast Charger 65W, Wireless Headphones
Fashion	Men Cotton T-Shirt, Running Shoes, Women Kurti
Grocery	Organic Honey 500g
Home & Kitchen	Mixer Grinder 750W, Non-stick Cookware Set, Steel Water Bottle 1L
Sports	Kashmir Willow Cricket Bat, Yoga Mat
Toys	Building Blocks Set

4.2 Number of products in each category

The query uses LEFT JOIN so that a category with no products would still appear with a count of 0.

Category_Name	Total_Products
Electronics	6
Fashion	3
Home & Kitchen	3
Beauty	2
Books	2
Sports	2
Grocery	1
Toys	1

4.3 Highest-priced product

The highest-priced product overall is the **Redmi Note 13 5G** (Electronics) at **16999.00**. The highest-priced product in each category is:

Category_Name	Product_Name	Price
Beauty	Moisturizer 200ml	349.00
Books	Atomic Habits (Book)	499.00
Electronics	Redmi Note 13 5G	16999.00
Fashion	Running Shoes	2299.00
Grocery	Organic Honey 500g	350.00
Home & Kitchen	Mixer Grinder 750W	3299.00
Sports	Kashmir Willow Cricket Bat	1299.00
Toys	Building Blocks Set	799.00

4.4 Categories having more than 5 products

The query groups the products by category and uses HAVING COUNT(product_id) > 5. HAVING is used because WHERE cannot filter on the result of COUNT.

Category_Name	Total_Products
Electronics	6

4.5 Average product price

The average price of all 20 products is **1931.60**. The average price in each category is:

Category_Name	Average_Price
Electronics	4182.33
Home & Kitchen	1865.67
Fashion	1165.67
Sports	999.00
Toys	799.00
Books	374.50
Grocery	350.00
Beauty	274.00

4.6 Category summary

Category_Name	Total_Products	Lowest_Price	Highest_Price	Average_Price	Total_Stock
Electronics	6	999.00	16999.00	4182.33	495
Fashion	3	499.00	2299.00	1165.67	380
Home & Kitchen	3	399.00	3299.00	1865.67	290
Beauty	2	199.00	349.00	274.00	440
Books	2	250.00	499.00	374.50	240
Sports	2	699.00	1299.00	999.00	115
Grocery	1	350.00	350.00	350.00	110

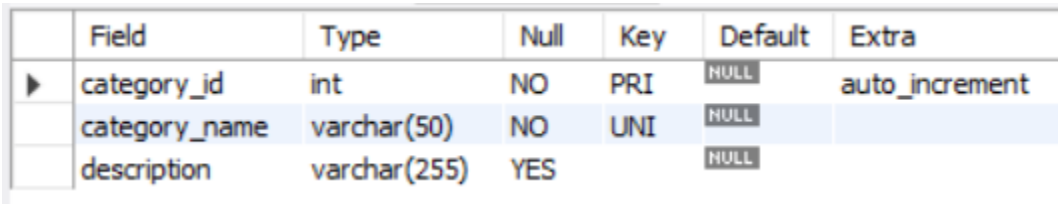
Category_Name	Total_Products	Lowest_Price	Highest_Price	Average_Price	Total_Stock
Toys	1	799.00	799.00	799.00	90

4.7 Key observations

- Electronics is the largest category, with 6 of the 20 products (30%). It is the only category with more than 5 products.
- Electronics also has the highest average price (4182.33) and the highest-priced product (Redmi Note 13 5G, 16999.00).
- The overall average of 1931.60 is pulled up by the Redmi Note 13 5G. The average of the other 19 products is about 1138.58.
- Beauty has the lowest average price (274.00) but the second-highest stock (440 units), so low-priced products are stocked in larger numbers.
- Toys and Grocery have only one product each, so they are the first categories to expand.
- The total stock across all categories is 2160 units, of which Electronics holds 495 (about 22.9%).

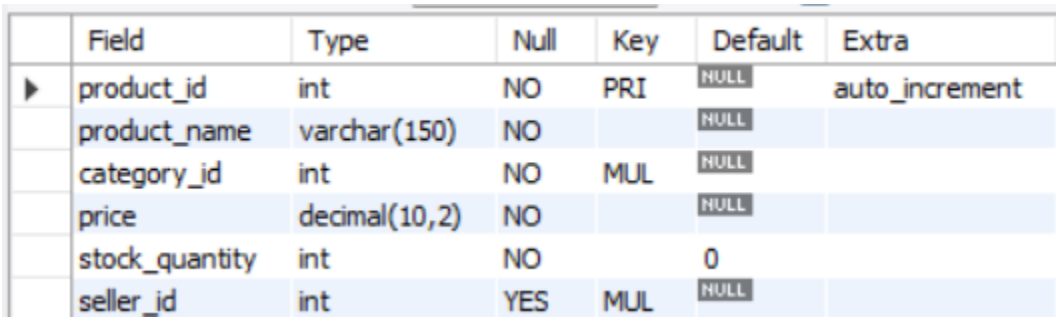
5. Query Output Screenshots

1. DESCRIBE category



Field	Type	Null	Key	Default	Extra
category_id	int	NO	PRI	NULL	auto_increment
category_name	varchar(50)	NO	UNI	NULL	
description	varchar(255)	YES		NULL	

2. DESCRIBE product



Field	Type	Null	Key	Default	Extra
product_id	int	NO	PRI	NULL	auto_increment
product_name	varchar(150)	NO		NULL	
category_id	int	NO	MUL	NULL	
price	decimal(10,2)	NO		NULL	
stock_quantity	int	NO		0	
seller_id	int	YES	MUL	NULL	

3. SELECT * FROM category

category_id	category_name	description
1	Electronics	Mobiles, audio devices and accessories
2	Fashion	Clothing and footwear for men and women
3	Home & Kitchen	Cookware, appliances and household items
4	Books	Fiction, non-fiction and self-help books
5	Sports	Sports equipment and fitness gear
6	Beauty	Skin care and personal care products
7	Toys	Toys and games for children
8	Grocery	Packaged food, staples and organic groceries
NULL	NULL	NULL

4. SELECT * FROM product

product_id	product_name	category_id	price	stock_quantity	seller_id
1	Redmi Note 13 5G	1	16999.00	50	1
2	Men Cotton T-Shirt	2	499.00	200	2
3	Non-stick Cookware Set	3	1899.00	80	3
4	Wings of Fire (Book)	4	250.00	150	4
5	Kashmir Willow Cricket Bat	5	1299.00	40	5
6	Wireless Headphones	1	1499.00	120	6
7	Mixer Grinder 750W	3	3299.00	60	7
8	Face Wash 100ml	6	199.00	300	8
9	Building Blocks Set	7	799.00	90	9
10	Organic Honey 500g	8	350.00	110	10
11	Bluetooth Speaker	1	1799.00	70	6
12	Smart Watch	1	2499.00	70	1
13	Power Bank 20000mAh	1	1299.00	100	6
14	USB-C Fast Charger 65W	1	999.00	85	1
15	Women Kurti	2	699.00	120	2
16	Running Shoes	2	2299.00	60	2
17	Steel Water Bottle 1L	3	399.00	150	3
18	Atomic Habits (Book)	4	499.00	90	4
19	Yoga Mat	5	699.00	75	5
20	Moisturizer 200ml	6	349.00	140	8
NULL	NULL	NULL	NULL	NULL	NULL

5. CRUD INSERT, then the SELECT with JOIN

```
INSERT INTO category (category_name, description)
VALUES ('Stationery', 'Pens, notebooks and office supplies');
SET @new_category_id = LAST_INSERT_ID();
```

```
36  22:33:14  INSERT INTO category (category_name, description) VALUES ('Stationery', 'Pens, notebooks and office suppli...
37  22:33:14  SET @new_category_id = LAST_INSERT_ID()
```

```
INSERT INTO product (product_name, category_id, price, stock_quantity, seller_id)
VALUES ('Gel Pen Pack (10)', @new_category_id, 120.00, 200, 4);
SET @new_product_id = LAST_INSERT_ID();
```

```
38  22:33:14  INSERT INTO product (product_name, category_id, price, stock_quantity, seller_id) VALUES ('Gel Pen Pack (...
39  22:33:14  SET @new_product_id = LAST_INSERT_ID()
```

```
SELECT p.product_id, p.product_name, c.category_name, p.price, p.stock_quantity
FROM product p
JOIN category c ON p.category_id = c.category_id
WHERE p.product_id = @new_product_id;
```

product_id	product_name	category_name	price	stock_quantity
23	Gel Pen Pack (10)	Stationery	120.00	200

6. SELECT * FROM product

```
WHERE product_id IN (11, 12)
```

product_id	product_name	category_id	price	stock_quantity	seller_id
11	Bluetooth Speaker	1	1799.00	70	6
12	Smart Watch	1	2499.00	95	1
NULL	NULL	NULL	NULL	NULL	NULL

7. DELETE FROM product WHERE product_id = @new_product_id;

```
81  23:07:54  DELETE FROM product  WHERE product_id  = @new_product_id
```

```
DELETE FROM category WHERE category_id = @new_category_id;
```

```
82  23:07:57  DELETE FROM category WHERE category_id = @new_category_id
```

8. **Products category-wise**

```
SELECT c.category_name, p.product_id, p.product_name, p.price, p.stock_quantity
FROM category c
JOIN product p ON c.category_id = p.category_id
ORDER BY c.category_name, p.product_name;
```

category_name	product_id	product_name	price	stock_quantity
Beauty	8	Face Wash 100ml	199.00	300
Beauty	20	Moisturizer 200ml	349.00	140
Books	18	Atomic Habits (Book)	499.00	90
Books	4	Wings of Fire (Book)	250.00	150
Electronics	11	Bluetooth Speaker	1799.00	70
Electronics	13	Power Bank 20000mAh	1299.00	100
Electronics	1	Redmi Note 13 5G	16999.00	50
Electronics	12	Smart Watch	2499.00	95
Electronics	14	USB-C Fast Charger 65W	999.00	85
Electronics	6	Wireless Headphones	1499.00	120
Fashion	2	Men Cotton T-Shirt	499.00	200
Fashion	16	Running Shoes	2299.00	60
Fashion	15	Women Kurti	699.00	120
Grocery	10	Organic Honey 500g	350.00	110
Home & Kitchen	7	Mixer Grinder 750W	3299.00	60
Home & Kitchen	3	Non-stick Cookware Set	1899.00	80
Home & Kitchen	17	Steel Water Bottle 1L	399.00	150
Sports	5	Kashmir Willow Cricket Bat	1299.00	40
Sports	19	Yoga Mat	699.00	75
Toys	9	Building Blocks Set	799.00	90

9. **Count of products in each category**

```
SELECT c.category_name, COUNT(p.product_id) AS total_products
FROM category c
LEFT JOIN product p ON c.category_id = p.category_id
GROUP BY c.category_id, c.category_name
ORDER BY total_products DESC, c.category_name;
```

category_name	total_products
Electronics	6
Fashion	3
Home & Kitchen	3
Beauty	2
Books	2
Sports	2
Grocery	1
Toys	1

10. Highest-priced product

```
SELECT p.product_id, p.product_name, c.category_name, p.price
FROM product p
JOIN category c ON p.category_id = c.category_id
WHERE p.price = (SELECT MAX(price) FROM product);
```

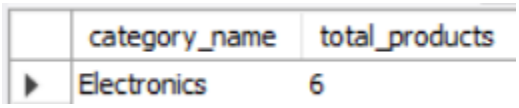
product_id	product_name	category_name	price
1	Redmi Note 13 5G	Electronics	16999.00

```
SELECT c.category_name, p.product_name, p.price
FROM product p
JOIN category c ON p.category_id = c.category_id
WHERE p.price = (SELECT MAX(p2.price) FROM product p2 WHERE p2.category_id = p.category_id)
ORDER BY c.category_name;
```

category_name	product_name	price
Beauty	Moisturizer 200ml	349.00
Books	Atomic Habits (Book)	499.00
Electronics	Redmi Note 13 5G	16999.00
Fashion	Running Shoes	2299.00
Grocery	Organic Honey 500g	350.00
Home & Kitchen	Mixer Grinder 750W	3299.00
Sports	Kashmir Willow Cricket Bat	1299.00
Toys	Building Blocks Set	799.00

11. Categories having more than 5 products

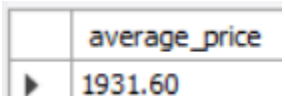
```
SELECT c.category_name, COUNT(p.product_id) AS total_products
FROM category c
JOIN product p ON c.category_id = p.category_id
GROUP BY c.category_id, c.category_name
HAVING COUNT(p.product_id) > 5;
```



	category_name	total_products
▶	Electronics	6

12. Average price

```
SELECT c.category_name, ROUND(AVG(p.price), 2) AS average_price
FROM category c
JOIN product p ON c.category_id = p.category_id
GROUP BY c.category_id, c.category_name
ORDER BY average_price DESC;
```



	average_price
▶	1931.60

13. Category summary

```
SELECT c.category_name,
    COUNT(p.product_id)         AS total_products,
    MIN(p.price)                AS lowest_price,
    MAX(p.price)                AS highest_price,
    ROUND(AVG(p.price), 2)      AS average_price,
    SUM(p.stock_quantity)       AS total_stock
FROM category c
JOIN product p ON c.category_id = p.category_id
GROUP BY c.category_id, c.category_name
ORDER BY total_products DESC, c.category_name;
```

category_name	total_products	lowest_price	highest_price	average_price	total_stock
Electronics	6	999.00	16999.00	4182.33	520
Fashion	3	499.00	2299.00	1165.67	380
Home & Kitchen	3	399.00	3299.00	1865.67	290
Beauty	2	199.00	349.00	274.00	440
Books	2	250.00	499.00	374.50	240
Sports	2	699.00	1299.00	999.00	115
Grocery	1	350.00	350.00	350.00	110
Toys	1	799.00	799.00	799.00	90

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